Locate all 19 original roller-control callouts

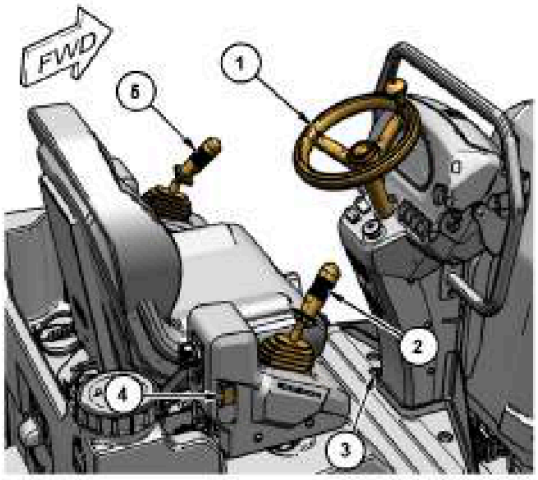


Illustration 39 g06673749

(1) Steering wheel
(2) Right propel control
(3) Traction Control Switch (if equipped)
(4) 12V power receptacle
(5) Left propel control (if equipped)

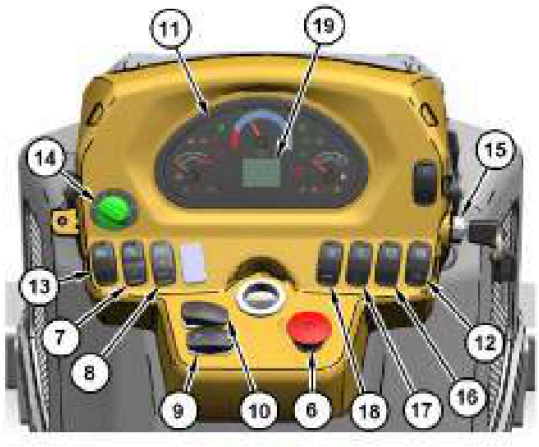


Illustration 40 g06610378

(6) Emergency stop knob
(7) Work lights
(8) Front/Position lights
(9) Turn signal
(10) Hazard lights
(11) Gauges and indicators
(12) Parking brake
(13) Horn
(14) Water spray system
(15) Engine start switch
(16) Vibratory mode / Emulsion switch
(17) Engine speed
(18) Beacon
(19) Digital display

Steering Wheel (1)

Steering wheel (1) controls the directional steering of the machine. The machine will turn in the same direction as steering wheel (1).

Note. Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 41). Faithful crop; original labels retained. CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Illustration 39 has 1–5; Illustration 40 has 6–19. Optional items retain “if equipped.” Full source context.

Worked example · A dry stripe on the drum

A drum shows a persistent dry stripe and material begins to pick up in that area.

Think it through.

Report the pattern, stop/hold as required and inspect the spray/scraper system through the safe procedure. A blocked nozzle or another fault is a possibility, not a confirmed diagnosis. Correct and verify the authorized repair before continuing.

Try it and explain your thinking

A fictional 120 L tank contains 35 L

A fictional 120 L tank contains 35 L. How much water reaches the stated 120 L level?

Skills you use: Numeracy, Reading.

List four water-system items for a model-specific winterization record.

List four water-system items for a model-specific winterization record.

Skills you use: Writing, Problem solving.

Build a maintenance checklist with AI assistance and audit it before use.

Build a maintenance checklist with AI assistance and audit it before use.

Skills you use: Digital, Communication, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Service lubrication/intake/separator and correct fluids safely.	0 / 1 / 2	Evidence/help: _____
Inspect cooling/battery and fitted safety systems.	0 / 1 / 2	Evidence/help: _____
Maintain sprays, scrapers/contact surfaces and model-specific winterization.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, L4, printed p. 97; source PDF pp. 99–100.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- Lincoln Industrial. (December 2013; Form 404636). Lever Gun Models 1142, 1147 and 1151: Installation and Maintenance Guide. Open publication.
- Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022)). CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline L5

Operate Compaction Roller

Compaction operation combines smooth machine control with temperature, density and coverage feedback. The instructor observes the actual machine and conditions; a written pattern alone does not demonstrate operating skill.

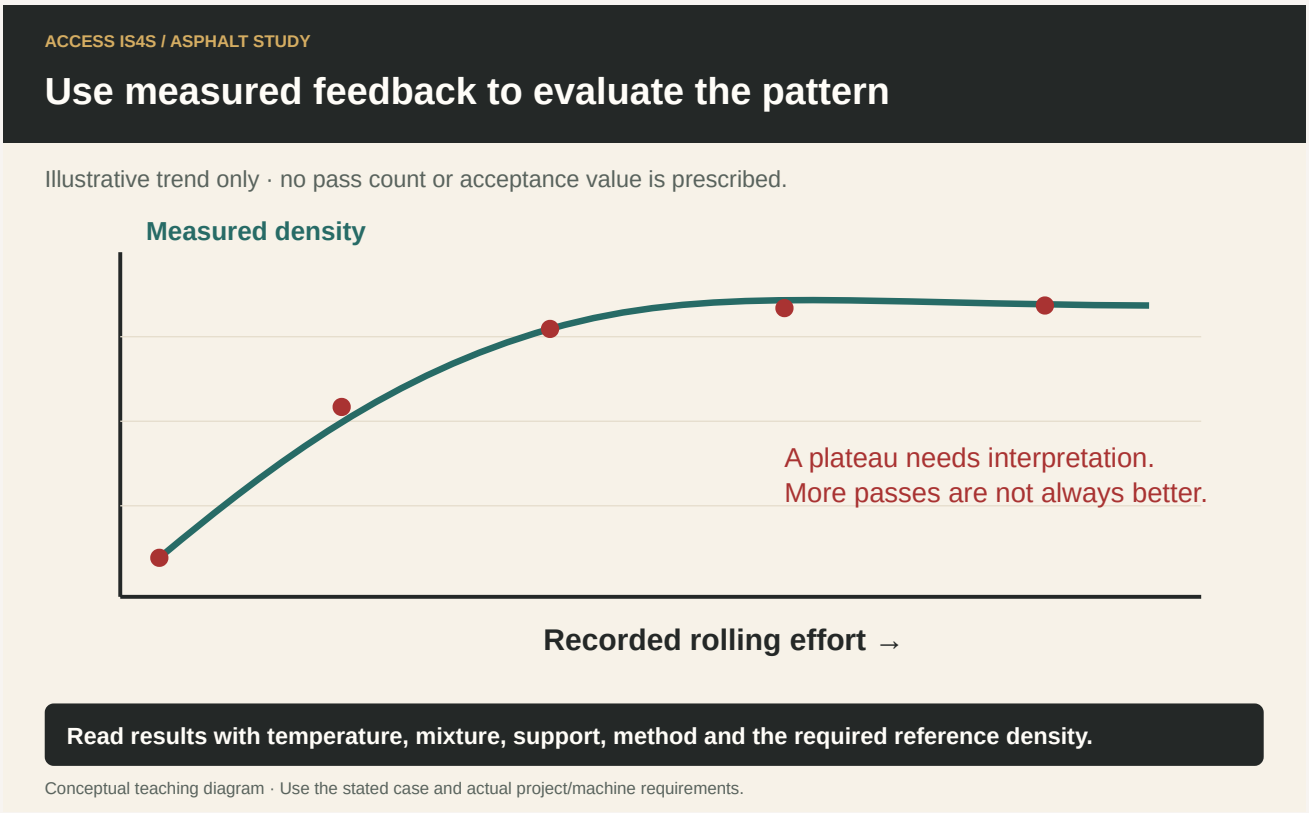
- Plan work
- Test temperature
- Test density
- Maintain efficient patterns

LEARNING STEP 1

Plan work

Inspect the grade for edges, support, obstacles and limits. Confirm roller types/roles, approved settings and pattern, water supply, communications and traffic controls. Complete pre-start checks and daily maintenance. Start and monitor through the matching OMM. Practise forward–stop–reverse–stop on the flat training area, parking controls, travel-mode changes and fitted vibration controls under supervision. Weather, site conditions, grade and quality feedback can require an authorized plan change.

Use measured feedback to evaluate the pattern



Note. ACCESS IS4S. (2026). Use measured feedback to evaluate the pattern [Original conceptual SVG]. Informed by Asphalt Institute (2007), The asphalt handbook, 7th ed., Chapter 9, including Fig. 9.9 (p. 435; PDF p. 469). New schematic trend; numerical axes and acceptance line omitted; no operating pattern prescribed.

LEARNING STEP 2

Test temperature

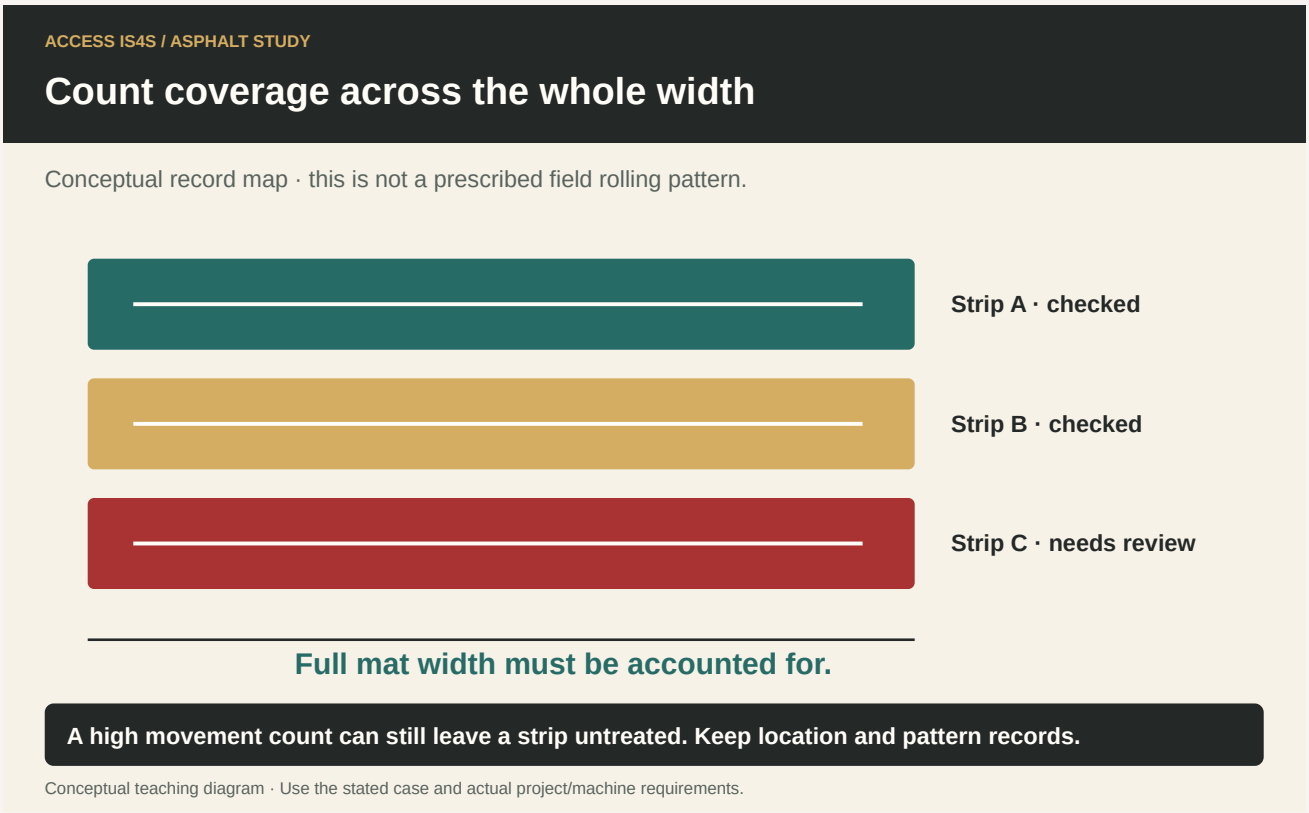
Measure temperature with the specified instrument and method at safe locations. Infrared instruments read a surface-related temperature and depend on emissivity, distance/spot size and conditions; a probe measures through its contact method. Record location, time, method and units. Mixture-specific compaction conditions govern; do not use a universal temperature window from memory. Cooling rate varies with lift, weather, support and delays.

LEARNING STEP 3

Test density

Density testing provides feedback through the approved method and reference. A nuclear gauge such as the Troxler 3430 requires licensed/trained use and applicable CNSC controls; observing it does not authorize operation. Core testing has a different sampling process. Confirm the denominator for percent density and the test location/lot. The test plan determines frequency. Results help the responsible lead evaluate the pattern; a single reading does not prove complete acceptance.

Count coverage across the whole width



Note. ACCESS IS4S. (2026). Count coverage across the whole width [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 4

Maintain efficient patterns

Maintain the approved breakdown, intermediate and finish sequence with smooth direction changes and controlled speed/mode. Demonstrate the assigned confined-edge and longitudinal-joint patterns under the instructor’s plan. Coordinate with the paver and other rollers, track coverage and avoid abrupt turns or damaging stops. Do not interpret repeated forward/reverse movement as permission to stop with vibration engaged. Finish with the model-specific shutdown and a clear record of actual treatment and unresolved areas.

Troxler 3430 gauge and accessories

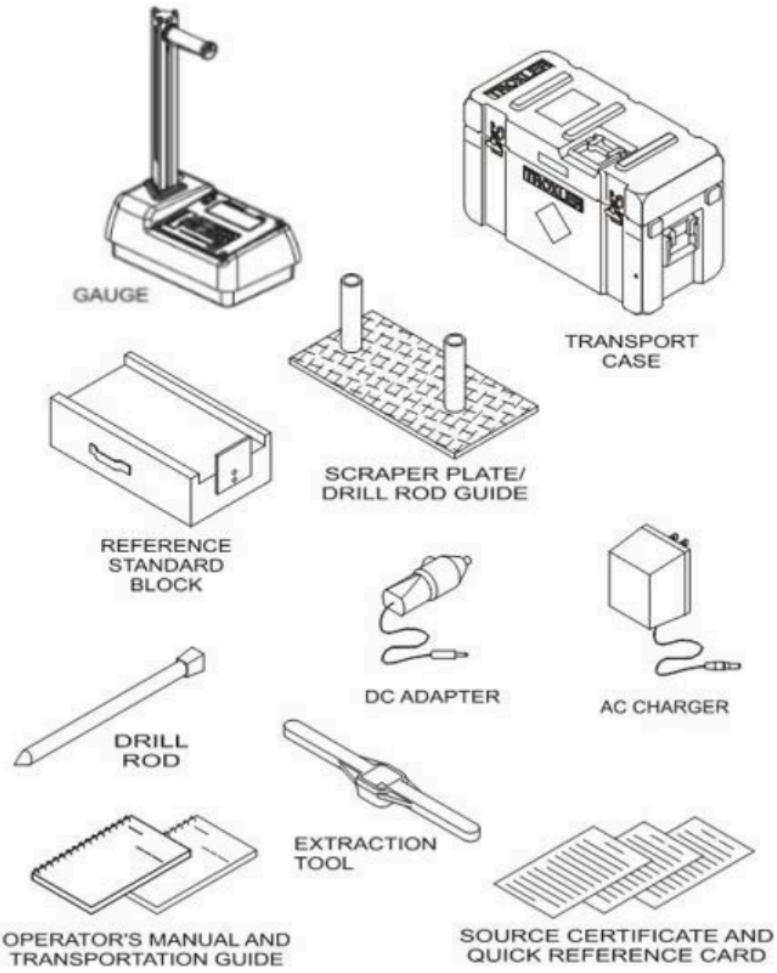


Figure 1. Model 3430 Gauge and Accessories

Note. Troxler Electronic Laboratories. (2021). RoadReader Model 3430: Manual of operation and instruction (Ed. 1.3, Fig. 1, p. 1-5; PDF p. 19). Identification only; Canadian use requires applicable CNSC authorization. Full source context.

Worked example · Density feedback changes the plan

A trial strip's recorded density increases over the first approved passes, then stops improving while the mixture cools.

Think it through.

Report the measured results with temperature, location and reference. The quality/crew lead evaluates whether settings, timing, equipment or another factor needs adjustment. More passes are not automatically effective and may damage the mat. Preserve the trial record and authorized revised pattern.

Try it and explain your thinking

In-place density is 2.310 Mg/m³ and the required maximum-theoretical reference is 2.500 Mg/m³

In-place density is 2.310 Mg/m³ and the required maximum-theoretical reference is 2.500 Mg/m³. Calculate percent.

Skills you use: Numeracy, Reading.

An infrared reading and probe reading differ

An infrared reading and probe reading differ. What should be checked before declaring either wrong?

Skills you use: Problem solving, Communication.

Create a rolling record that another operator can continue from.

Create a rolling record that another operator can continue from.

Skills you use: Digital, Writing, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Precheck, service, start/monitor and shut down the actual roller.	0 / 1 / 2	Evidence/help: _____
Perform flat forward/stop/reverse/stop, parking and travel-mode controls safely.	0 / 1 / 2	Evidence/help: _____
Operate fitted vibration through the approved sequence.	0 / 1 / 2	Evidence/help: _____
Demonstrate assigned confined-edge and longitudinal-joint patterns.	0 / 1 / 2	Evidence/help: _____
Coordinate breakdown/intermediate/finish roles with the paving train.	0 / 1 / 2	Evidence/help: _____
Interpret temperature/density feedback and maintain a traceable coverage record.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, L5, printed p. 99; source PDF pp. 101–102.
- Caterpillar. (2012). Guide to Asphalt Compaction (QEBQ1583). Asphalt compaction concepts and equipment; project and machine application require the applicable documents. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- ThermoWorks. (n.d.). Industrial IR with Circle Laser: IR-IND / IRK-2 Operating Instructions. Open publication.
- Troxler Electronic Laboratories. (Edition 1.3, February 2021). RoadReader Model 3430 Surface Moisture-Density Gauge: Manual of Operation and Instruction. Open publication.
- Canadian Nuclear Safety Commission. (Web snapshot: October 30, 2026). Portable Nuclear Gauges. Open publication.
- Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022)). CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline L6

Follow shut down procedures

A roller can move unintentionally after the shift if parking and securement are incomplete. Shutdown also prepares the machine for cleaning, staging, maintenance and transport.

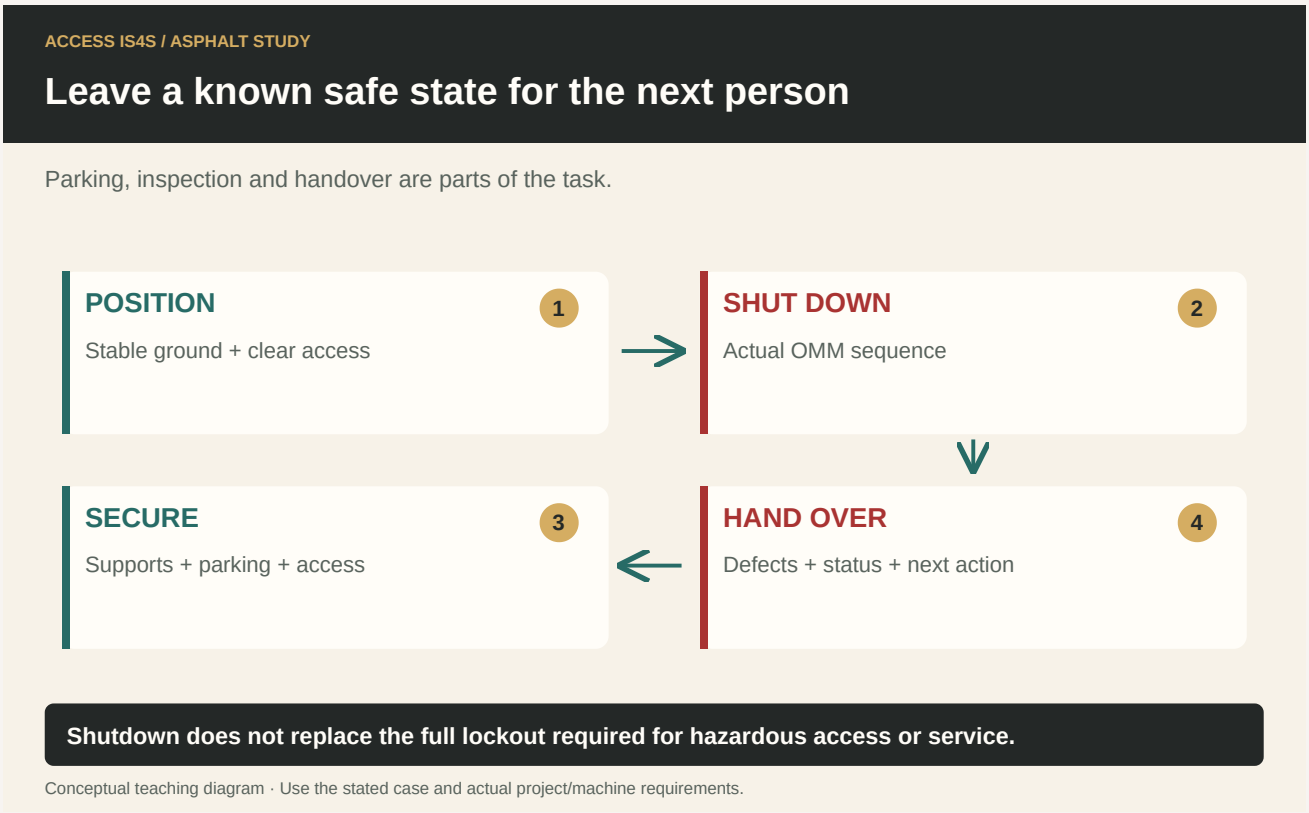
- Shut-down procedures
- Clean equipment
- Ensure efficient staging
- Prepare for transport
- Secure machine for safety

LEARNING STEP 1

Shut-down procedures

Choose the designated stable parking position with service access, clear emergency/public routes and suitable ground for checks. Follow the OMM stopping, cooling and shutdown sequence, apply parking controls and required chocks, and inspect safely for leaks or damage. Use the battery/night switch only as directed; it is not the whole lockout procedure. Record defects and the machine’s status.

Leave a known safe state for the next person



Note. ACCESS IS4S. (2026). Leave a known safe state for the next person [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Clean equipment

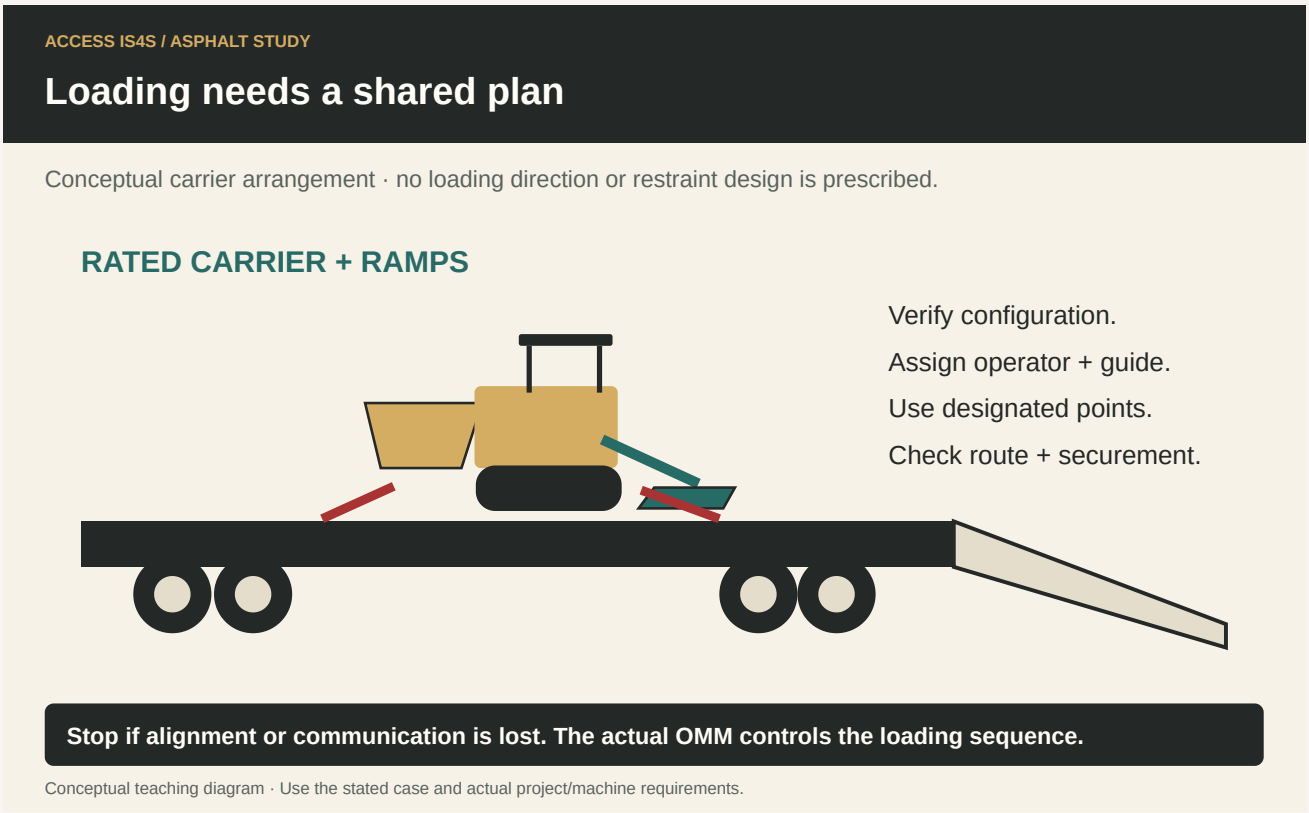
Clean drums/tires, scrapers and spray equipment through the approved safe method. Control movement, heat, pressure and cleaning-product hazards, and contain waste. Never reach toward a rotating drum or articulation zone to remove pickup. Restore guards and account for tools before release.

LEARNING STEP 3

Ensure efficient staging

Coordinate staging with the foreperson so the next shift can access the roller without blocking other equipment, deliveries or public routes. Identify the machine, location, readiness and pending service. Efficient staging includes safe access for a machine that cannot restart.

Loading needs a shared plan



Note. ACCESS IS4S. (2026). Loading needs a shared plan [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 4

Prepare for transport

Prepare for transport using the actual OMM configuration and carrier plan. Assign a competent operator/guide and carrier responsibilities, verify mass/dimensions, ramp/support suitability and designated tie-down points. Use the applicable securement requirements and approved loading sequence. The transport driver’s role does not remove the need for clear machine-control responsibilities.

LEARNING STEP 5

Secure machine for safety

Secure controls, keys, detachable items and access panels; protect the public and prevent roll-away through the required parking/chocking arrangement. Consider residual heat and stored energy. Communicate out-of-service status visibly through the site system. Recheck the final state before leaving; a completed numerical checklist score cannot override a failed critical securement item.

Keep the contact surface working consistently



Note. ACCESS IS4S. (2026). Keep the contact surface working consistently [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Ready for the next shift

A roller is clean and fuelled but parked where a delivery truck must reverse around it beside a public entrance.

Think it through.

Have the responsible lead arrange a safer approved staging location. Complete the movement and shutdown under the actual procedure, preserve access and communicate the final location/status. “Ready” includes safe placement, not only clean equipment.

Try it and explain your thinking

State the difference between staging and transport securement.

State the difference between staging and transport securement.

Skills you use: Reading, Communication.

A ten-item checklist has 20 possible points

A ten-item checklist has 20 possible points. A learner earns 18 but fails a critical roll-away control. Is the outcome satisfactory?

Skills you use: Numeracy, Problem solving.

Write a final shift handover with a pending spray-system repair.

Write a final shift handover with a pending spray-system repair.

Skills you use: Writing, Digital, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 4 task criteria out of 8; require at least 7 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Perform matching shutdown, parking and required chocking.	0 / 1 / 2	Evidence/help: _____
Clean safely and stage without blocking access.	0 / 1 / 2	Evidence/help: _____
Prepare transport configuration and communicate roles.	0 / 1 / 2	Evidence/help: _____
Secure the machine and hand over defects/readiness.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, L6, printed p. 101; source PDF pp. 103–103.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022)). CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Integrated practice · all nine Skills for Success

Read the rolling record and its limits

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional density record gives in-place 2.325 Mg/m³ and maximum theoretical 2.500 Mg/m³. A coverage sketch marks strips A and B complete and C awaiting a check. A draft says “93% means the whole lot is accepted” and “six passes always work.” No contract limit, sub-lot results or approved fixed pass count is supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Change card · Use after the first attempt

The spray system develops a dry stripe and the operator observes material pickup.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person’s actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.